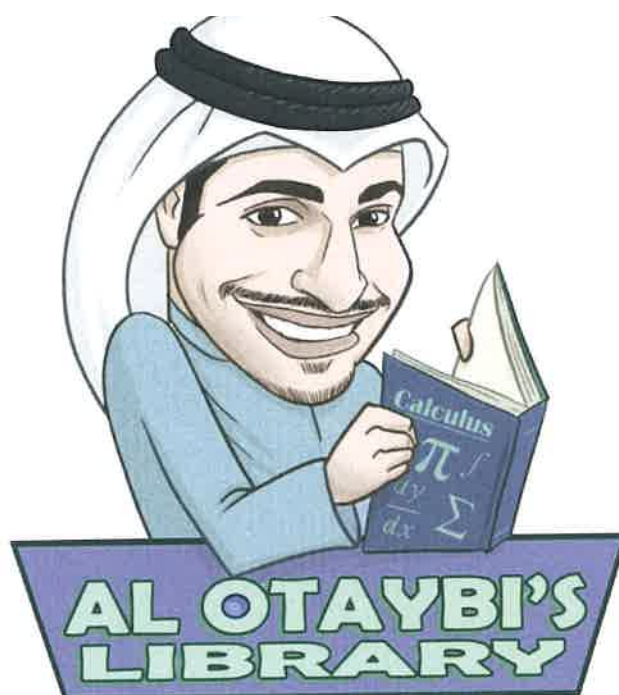


Kuwait University

2021/2022

Fall course



Calculus A

First

Sections : 3.1 – 3.2

3.1 Derivatives of Polynomials and Exponential Functions.

3.2 The Product and Quotient Rules.

(9)



3.1 Derivative of Polynomials and Exponential Functions



$$(x^n)' = nx^{n-1}$$



Differentiate the function

$$f(x) = x^3$$

$$f'(x) = 3x^2$$

$$f(x) = x^4$$

$$f'(x) = 4x^3$$

$$f(x) = x^{-3}$$

$$f'(x) = -3x^{-4}$$

$$f(x) = x^{\frac{3}{2}}$$

$$f'(x) = \frac{3}{2} x^{\frac{1}{2}}$$

$$f(x) = x^e$$

$$f'(x) = e x^{e-1}$$

$$f(x) = x^\pi$$

$$f'(x) = \pi x^{\pi-1}$$



أختصارات حلوة



$$(\sqrt{x})' = \frac{1}{2\sqrt{x}}$$

$$\left(\frac{1}{x}\right)' = \frac{-1}{x^2}$$

Differentiate the function

$$f(x) = \sqrt{x^3 + x^2}$$

$$f'(x) = \frac{3x^2 + 2x}{2\sqrt{x^3 + x^2}}$$



مشتقة الرقم



بجانبه x



أنزله كما هو
ثم أشتق الي يمه

بروحه



0

Differentiate the function

$$f(x) = 5$$

$$f'(x) = 0$$

$$f(x) = 8x^2$$

$$f'(x) = 8(2x) = 16x$$

$$f(x) = 8$$

$$f'(x) = 0$$

$$f(x) = 3\sqrt{x}$$

$$f'(x) = 3 \frac{1}{2\sqrt{x}}$$

$$f(x) = e$$

$$f'(x) = 0$$

$$f(x) = \frac{3}{x}$$

$$f'(x) = \frac{-3}{x^2}$$



Text book question



Differentiate the function

$$f(x) = 5.2x + 2.3$$

$$f'(x) = 5.2(1) + 0$$

$$y = x^{5/3} - x^{2/3}$$

$$y' = \frac{5}{3}x^{2/3} - \frac{2}{3}x^{-1/3}$$

$$f(t) = 2t^3 - 3t^2 - 4t$$

$$f'(t) = 6t^2 - 6t - 4$$

$$g(x) = x^2(1 - 2x)$$

$$g(x) = x^2 - 2x^3$$

$$g'(x) = 2x - 6x^2$$



$$(e^x)' = e^x$$

$$(e^{f(x)})' = e^{f(x)} f'(x)$$



Differentiate the function

$$f(x) = e^x$$

$$f'(x) = e^x (1)$$

$$f(x) = e^{4x}$$

$$f'(x) = e^{4x} (4)$$

$$f(x) = e^{x^2}$$

$$f'(x) = e^{x^2} (2x)$$



$$f(x) = e^{x^2} + e^{x^3+1} + e^4 + e^x + ex - x^e - \sqrt{x^2 + 1}$$



$$f'(x) = e^{x^2} (2x) + e^{x^3+1} (3x^2) + 0 + e^x + e(1) - ex^{e-1} - \frac{2x}{2\sqrt{x^2+1}}$$



Text book question



Differentiate the function

$$f(r) = \frac{5}{r^3}$$

$$f(r) = 5 r^{-3}$$

$$f'(r) = -15 r^{-4}$$

$$y = 3e^x + \frac{4}{\sqrt[3]{x}}$$

$$y = 3e^x + 4x^{-\frac{1}{3}}$$

$$y' = 3e^x + 4(-\frac{1}{3}x^{-\frac{4}{3}})$$

$$j(x) = x^{2.4} + e^{2.4}$$

$$j'(x) = 2.4x^{1.4} + 0$$

$$y = ae^v + \frac{b}{v} + \frac{c}{v^2}$$

$$y = ae^v + \frac{b}{v} + c v^{-2}$$

$$y' = ae^v + \frac{-b}{v^2} + c(-2v^{-3})$$



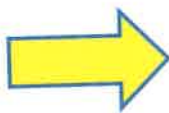
$$f'$$



tangent line – normal line – slope
increasing – decreasing – critical numbers

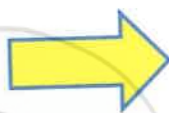


Slope of the **tangent** line



$$m_t = f'(a)$$

Slope of the **normal** line



$$m_n = \frac{1}{f'(a)}$$

tangent line has slope a



$$f'(x) = a$$

Equation of the **tangent** line



$$y - y_o = m_t(x - x_o)$$

Equation of the **normal** line



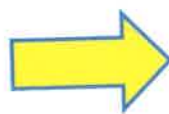
$$y - y_o = m_n(x - x_o)$$

Tangent line is **horizontal**



$$f'(x) = 0$$

Tangent line is **vertical**



$$f'(x) \text{ undefind}$$



الممنوعات



(1) ممنوع أس سالب
(2) ممنوع مقام غير موحد

1) f is continuous at $x = a$

$$2) \lim_{x \rightarrow a} |f'(x)| = \infty$$



Text book questions



Find an equation of tangent line to the curve at the given point.

$$y = 2x^3 - x^2 + 2, (1, 3)$$

$$y' = 6x^2 - 2x$$

$$\text{at } (1, 3) \longrightarrow y' = 6(1)^2 - 2(1)$$

$$y' = 6 - 2 \longrightarrow$$

$$y' = 4$$

∴ The equation of the tangent line :

$$y - y_0 = m(x - x_0)$$

$$y - 3 = 4(x - 1)$$

Find an equation of tangent line to the curve at the given point.

$$y = 2e^x + x, (0, 2)$$

$$y' = 2e^x + 1$$

$$\text{at } (0, 2) \longrightarrow y' = 2e^0 + 1$$

$$y' = 2 + 1 \longrightarrow$$

$$y' = 3$$

∴ The equation of the tangent line :

$$y - y_0 = m(x - x_0)$$

$$y - 2 = 3(x - 0)$$



Text book questions



Find an equation of tangent line to the curve at the given point.

$$y = x + \frac{2}{x}, (2, 3)$$

$$y' = 1 + \frac{-2}{x^2}$$

$$\text{at } (2, 3) \rightarrow y' = 1 + \frac{-2}{(2)^2}$$

$$y' = 1 - \frac{2}{4} \rightarrow y' = \frac{2}{2} - \frac{1}{2} \rightarrow y' = \frac{1}{2}$$

$\therefore$ The equation of the tangent line:

$$y - y_0 = m(x - x_0)$$

$$y - 3 = \frac{1}{2}(x - 2)$$

Find an equation of tangent line to the curve

$$y = \sqrt[4]{x} - x, \text{ at } x = 1$$

$$y = x^{\frac{1}{4}} - x$$

$$\text{at } x = 1 \rightarrow y = 1 - 1 = 0 \rightarrow P(1, 0)$$

$$y' = \frac{1}{4}x^{-\frac{3}{4}} - 1 = \frac{1}{4x^{\frac{3}{4}}} - 1$$

$$\text{at } (1, 0) \rightarrow y' = \frac{1}{4} - 1$$
$$y' = \frac{1}{4} - \frac{4}{4} \rightarrow y' = \frac{-3}{4}$$

$\therefore$ The equation of the tangent line:

$$y - y_0 = m(x - x_0)$$

$$y - 0 = \frac{-3}{4}(x - 1)$$



Text book questions



For what value of x does the graph of $f(x) = e^x - 2x$ have a horizontal tangent.

$$f'(x) = e^x - 2$$

∴ The tangent line is horizontal

$$∴ f'(x) = 0$$

$$e^x - 2 = 0$$

$$e^x = 2$$

$$\ln e^x = \ln 2$$

$$x = \ln 2$$



Extra



For what

point

$$\text{at } x = \ln 2$$

$$y = e^{\ln 2} - 2 \ln 2$$

$$y = 2 - 2 \ln 2$$

$$\Rightarrow \therefore P(\ln 2, 2 - 2 \ln 2)$$

Let $f(x) = e^x - 2x$

a) Find the point on the curve of f where the tangent line is horizontal.

b) Using part (a) find an equation of this horizontal tangent line.

Nov 2017

$$f'(x) = e^x - 2$$

∴ The tangent line is horizontal

$$∴ f'(x) = 0$$

$$e^x - 2 = 0$$

$$e^x = 2$$

$$\ln e^x = \ln 2 \longrightarrow x = \ln 2$$

$$\text{at } x = \ln 2 \longrightarrow y = e^{\ln 2} - 2\ln 2 = 2 - 2\ln 2$$
$$\implies ∴ P(\ln 2, 2 - 2\ln 2)$$

(b) The equation of the tangent line

$$y = 2 - 2\ln 2$$

Find the points on the curve of $y = x^3 - x^2 - x$

Where the tangent line is horizontal (parallel to the x-axis)

$$y = x^3 - x^2 - x$$

$$y' = 3x^2 - 2x - 1$$

∴ tangent line is horizontal

$$y' = 0$$

$$3x^2 - 2x - 1 = 0$$

$$(3x + 1)(x - 1) = 0$$

$$3x + 1 = 0$$

$$x - 1 = 0$$

$$x = -\frac{1}{3}$$

$$x = 1$$

$$y = \left(-\frac{1}{3}\right)^3 - \left(-\frac{1}{3}\right)^2 - \left(-\frac{1}{3}\right)$$

$$= -\frac{1}{27} - \frac{1}{9} + \frac{1}{3}$$

$$= -\frac{1}{27} - \frac{3}{27} + \frac{9}{27}$$

$$= \frac{5}{27}$$

$$y = (1)^3 - (1)^2 - 1$$

$$= 1 - 1 - 1 = -1$$

$$\Rightarrow \left(-\frac{1}{3}, \frac{5}{27}\right) \quad (1, -1)$$



Text book question



Show that the curve $y = 2e^x + 3x + 5x^3$ has no tangent line with slope 2.

$$y' = 2e^x(1) + 3(1) + 15x^2$$

$$y' = 2e^x + 3 + 15x^2$$

$$\Rightarrow y' = 2$$

$$2e^x + 3 + 15x^2 = 2 \quad (\text{which is impossible})$$

$\downarrow$ $\downarrow$
 70 70

$$\therefore 2e^x + 3 + 15x^2 > 3$$

$\Rightarrow \therefore y$ has no tangent line with slope 2



Do Not forget



زوی

$$\sqrt{x}$$

زوی

$$|x|$$

نوائجهم

$$70$$

e^a

a^x

نوائجهم

$$70$$



$$f'$$



tangent line – normal line – slope
increasing – decreasing – critical numbers



If we have two curves



Tangent line of $f(x)$ is

parallel

to the line y

Tangent line of $f(x)$ is

**Perpendicular –
orthogonal – meet at
right angle.**

to the line y



$$f'(x) = y'$$



$$f'(x) = -\frac{1}{y'}$$



The same



خطوة بالاثباتات

$$f'(x) \cdot y' = -1$$



Text book question



Find an equation of the tangent line to the curve

$y = x^4 + 1$ that is parallel to the line $32x - y = 15$

$$y = x^4 + 1$$

$$y'_1 = 4x^3$$

$$32x - y = 15$$

$$y = 32x - 15$$

$$y'_2 = 32$$

∴ The tangent line of y is parallel to the line

$$∴ y'_1 = y'_2$$

$$4x^3 = 32$$

÷ 4

$$x^3 = 8$$

$$x = 2$$

$$\text{at } x=2 \longrightarrow y' = 2^4 + 1 = 16 + 1 \longrightarrow ∴ y = 17$$



$$P(2, 17)$$

∴ The equation of the tangent line :

$$y - y_0 = m(x - x_0)$$

$$y - 17 = 32(x - 2)$$

Find the points on the graph of f at which the tangent line is perpendicular to the line $7y + x = 1$

where $f(x) = x^3 + x^2 + 6x - 5$

$$f(x) = x^3 + x^2 + 6x - 5$$

$$f'(x) = 3x^2 + 2x + 6$$

∴ The tangent line of f is perpendicular to the line y

$$\therefore f'(x) = -\frac{1}{y'}$$

$$3x^2 + 2x + 6 = 7$$

$$3x^2 + 2x - 1 = 0$$

$$(3x - 1)(x + 1) = 0$$

$$3x - 1 = 0$$

$$x + 1 = 0$$

$$3x = 1$$

$$x = -1$$

÷ 3

$$x = \frac{1}{3}$$

$$\begin{aligned} \text{at } x = \frac{1}{3} \rightarrow y &= \left(\frac{1}{3}\right)^3 + \left(\frac{1}{3}\right)^2 + 6\left(\frac{1}{3}\right) - 5 \\ &= \frac{1}{27} + \frac{1}{9} + 2 - 5 \\ &= \frac{1}{27} + \frac{1}{9} - 3 \end{aligned}$$

$$\begin{aligned} \text{at } x = -1 \rightarrow y &= -1 + 1 - 6 - 5 \\ &= -11 \end{aligned}$$

$$\Rightarrow P\left(\frac{1}{3}, \frac{1}{27} + \frac{1}{9} - 3\right), P(-1, -11)$$

$$7y + x = 1$$

$$7y = 1 - x$$

÷ 7

$$y = \frac{1}{7}(1 - x)$$

$$y = \frac{1}{7} - \frac{1}{7}x$$

$$y' = 0 - \frac{1}{7} \quad ||$$

$$y' = -\frac{1}{7}$$

Let $f(x) = ax^2 - 12x + 8$

Find all values of a such that the tangent line to the graph of f at $x = 2$ is perpendicular to the line $y + \frac{x}{4} + 1 = 0$

$$f(x) = ax^2 - 12x + 8$$

$$f'(x) = 2ax - 12$$

at $x = 2$

$$f'(2) = 2a(2) - 12$$
$$= 4a - 12$$

$$y + \frac{x}{4} + 1 = 0$$

$$y = -\frac{1}{4}x - 1$$

$$y' = -\frac{1}{4}(1) = -\frac{1}{4}$$

$$y' = -\frac{1}{4}$$

∴ The tangent line of f is perpendicular to the line y

$$\therefore f'(2) = -\frac{1}{y'}$$

$$4a - 12 = 4$$

$$4a = 16$$

$$\div 4$$

$$a = 4$$



Tricky question !!



Text book question



Find an equation of the normal line to the parabola

$y = x^2 - 5x + 4$ that is parallel to the line $x - 3y = 5$

$$y = x^2 - 5x + 4$$

$$y'_1 = 2x - 5$$

∴ The normal line is parallel to the line.

$$∴ y'_1 = -\frac{1}{y'_2}$$

$$2x - 5 = -3$$

$$\div 2 \quad 2x = 2$$

$$x = 1$$

$$y = 1 - 5 + 4 = 0$$

$$y = 0$$

$$\Rightarrow P(1, 0)$$

∴ Equation of the normal line :

$$y - y_0 = m(x - x_0)$$

$$y - 0 = \frac{1}{3}(x - 1)$$

$$x - 3y = 5$$

$$x - 5 = 3y$$

$$y = \frac{1}{3}(x - 5)$$

$$y = \frac{1}{3}x - \frac{5}{3}$$

$$y'_2 = \frac{1}{3}(1)$$

AL OTAYBI'S LIBRARY



Text book question



Find an equation of the normal line to the curve

$y = \sqrt{x}$ that is parallel to the line $2x + y = 1$

$$y = \sqrt{x}$$

$$y' = \frac{1}{2\sqrt{x}}$$

$$2x + y = 1$$

$$y = 1 - 2x$$

$$y' = -2$$

∴ The Normal line of y is parallel to the Line

$$\therefore y'_1 = \frac{-1}{y'_2}$$

$$\frac{1}{2\sqrt{x}} = \frac{1}{2}$$

$$2 = 2\sqrt{x}$$

÷ 2

$$1 = \sqrt{x}$$

$$x = 1$$

$$\text{at } x = 1 \longrightarrow y = \sqrt{1} = 1$$

$$\Rightarrow \boxed{P(1,1)}$$

∴ The equation of the normal line :

$$y - y_0 = m(x - x_0)$$

$$y - 1 = -2(x - 1)$$



Text book question



Evaluate the limit if it exist

$$\lim_{x \rightarrow 1} \frac{x^{1000} - 1}{x - 1}$$

$$\frac{1-1}{1-1} = \frac{0}{0}$$

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

$$a = 1$$

$$f(x) = x^{1000}$$

$$f'(x) = 1000 x^{999}$$

$$f'(a) = f'(1) = 1000 (1)^{999} = 1000$$

$$\Rightarrow \therefore \lim_{x \rightarrow 1} \frac{x^{1000} - 1}{x - 1} = 1000$$



Text book question



Let

$$f(x) = \begin{cases} x^2 + 1 & \text{if } x < 1 \\ x + 1 & , x \geq 1 \end{cases}$$

Is f differentiable at 1 ?

$$f'(a^+) = \lim_{x \rightarrow a^+} \frac{f(x) - f(a)}{x - a}$$

$$f'(a^-) = \lim_{x \rightarrow a^-} \frac{f(x) - f(a)}{x - a}$$

$$f'(1^+) = \lim_{x \rightarrow 1^+} \frac{x+1 - [1+1]}{x-1}$$

$$f'(1^-) = \lim_{x \rightarrow 1^-} \frac{x^2 + 1 - [1+1]}{x-1}$$

$$= \lim_{x \rightarrow 1^+} \frac{x+1-2}{x-1}$$

$$= \lim_{x \rightarrow 1^-} \frac{x^2 + 1 - 2}{x-1}$$

$$= \lim_{x \rightarrow 1^+} \frac{\cancel{x-1}}{\cancel{x-1}}$$

$$= \lim_{x \rightarrow 1^-} \frac{x^2 - 1}{x-1}$$

$$= \boxed{1}$$

$$= \lim_{x \rightarrow 1^-} \frac{(x-1)(x+1)}{\cancel{x-1}}$$

$$= 1+1 = \boxed{2}$$

$$\therefore f'(1^+) \neq f'(1^-)$$

$\therefore$ at $x=1 \rightarrow f$ is not differentiable.



Text book question



At what number is the following function g differentiable ?

$$f(x) = \begin{cases} 2x & \text{if } x \leq 0 \\ 2x - x^2 & \text{if } 0 < x < 2 \\ 2 - x & \text{if } x \geq 2 \end{cases}$$

$$f'(a^+) = \lim_{x \rightarrow a^+} \frac{f(x) - f(a)}{x - a}$$

$$f'(0^+) = \lim_{x \rightarrow 0^+} \frac{2x - x^2 - 0}{x - 0}$$

$$= \lim_{x \rightarrow 0^+} \frac{x(2-x)}{x}$$

$$= 2 - 0 = \boxed{2}$$

$$f'(a^-) = \lim_{x \rightarrow a^-} \frac{f(x) - f(a)}{x - a}$$

$$f'(0^-) = \lim_{x \rightarrow 0^-} \frac{2x - 0}{x - 0}$$

$$= \lim_{x \rightarrow 0^-} \frac{2x}{x}$$

$$= \boxed{2}$$

$\therefore f'(0^+) = f'(0^-) \longrightarrow \therefore$ at $x=0$ f is differentiable

$$f'(a^+) = \lim_{x \rightarrow a^+} \frac{f(x) - f(a)}{x - a}$$

$$f'(2^+) = \lim_{x \rightarrow 2^+} \frac{2 - x - (0)}{x - 2}$$

$$= \lim_{x \rightarrow 2^+} \frac{-(x-2)}{x-2}$$

$$= \boxed{-1}$$

$$f'(a^-) = \lim_{x \rightarrow a^-} \frac{f(x) - f(a)}{x - a}$$

$$f'(2^-) = \lim_{x \rightarrow 2^-} \frac{2x - x^2 - (0)}{x - 2}$$

$$= \lim_{x \rightarrow 2^-} \frac{-x(x-2)}{x-2}$$

$$= \boxed{-2}$$

$\therefore f'(2^+) \neq f'(2^-) \longrightarrow \therefore$ at $x=2$ f is not differentiable.

$\Rightarrow \therefore f$ is differentiable on $\mathbb{R} \setminus \{2\}$



Text book question



Let

$$f(x) = \begin{cases} x^2 & \text{if } x \leq 2 \\ mx + b & , x > 2 \end{cases}$$

Find the values of m and b that make f differentiable everywhere

$$\begin{aligned} f'(a^+) &= \lim_{x \rightarrow a^+} \frac{f(x) - f(a)}{x - a} \\ &= \lim_{x \rightarrow 2^+} \frac{mx + b - [2m + b]}{x - 2} \\ &= \lim_{x \rightarrow 2^+} \frac{mx + b - 2m - b}{x - 2} \\ &= \lim_{x \rightarrow 2^+} \frac{m(x - 2)}{x - 2} \\ &= \boxed{m} \end{aligned}$$

$$\begin{aligned} f'(a^-) &= \lim_{x \rightarrow a^-} \frac{f(x) - f(a)}{x - a} \\ &= \lim_{x \rightarrow 2^-} \frac{x^2 - 4}{x - 2} \\ &= \lim_{x \rightarrow 2^-} \frac{(x-2)(x+2)}{x-2} \\ &= 2 + 2 = \boxed{4} \end{aligned}$$

$\therefore f$ is differentiable everywhere $\rightarrow \therefore f$ is differentiable at $x=2$

$$\therefore \boxed{m = 4}$$

at $x=2$

$$f(2) = \boxed{4}$$

$$\lim_{x \rightarrow 2^+} (mx + b) = \boxed{2m + b}$$

$$\lim_{x \rightarrow 2^-} x^2 = \boxed{4}$$

$\therefore f$ is differentiable at $x=2$

$\therefore f$ is continuous at $x=2$

$$\therefore f(2) = \lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^-} f(x)$$

$$4 = 2m + b = 4$$

$$2m + b = 4$$

$$2(4) + b = 4$$

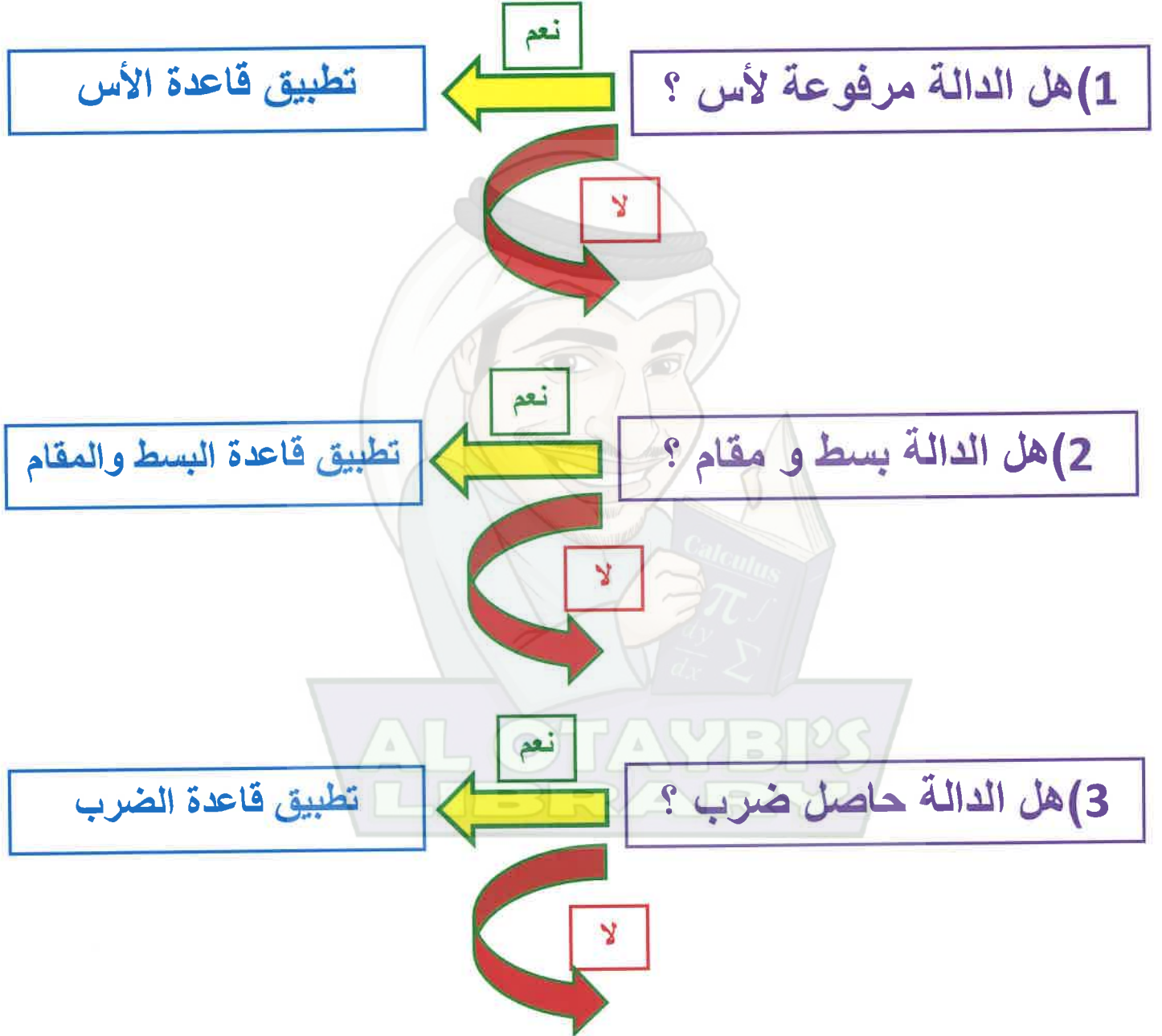
$$8 + b = 4$$

$$\rightarrow \boxed{b = -4}$$

3.2 The Product and Quotient Rules



طريقة اشتقاق أي دالة



الاشتقاق البسيط



Differentiate

$$f(x) = [1 + 2x^3]^5$$

$$f'(x) = 5 [1 + 2x^3]^4 \cdot 6x^2$$

اشتق



زلقة بطيخة !

$$f(x) = \frac{1 + 2x}{3 - 4e^x}$$

$$f'(x) = \frac{\text{مشتقة المقام} \cdot (\text{البسط}) - (\text{مشتقة البسط}) \cdot (\text{المقام})}{(\text{المقام})^2}$$

$$f'(x) = \frac{(3 - 4e^x)(0 + 2) - (1 + 2x)(-4e^x)}{(3 - 4e^x)^2}$$

$$f(x) = (1 + e^x)(2x^3 - 7)$$

$$f'(x) = (\text{اشتق}) \cdot (\text{ثبت}) + (\text{ثبت}) \cdot (\text{اشتق})$$

$$f'(x) = (0 + e^x)(2x^3 - 7) + (1 + e^x)(6x^2 - 0)$$



Text book questions



Differentiate

$$f(x) = (3x^2 - 5x)e^x$$

استق بـ + بـ استق

$$f'(x) = (6x - 5)e^x + (3x^2 - 5x)e^x \quad (1)$$

$$g(x) = (x + 2\sqrt{x})e^x$$

استق بـ + بـ استق

$$g'(x) = \left(1 + 2 \cdot \frac{1}{2\sqrt{x}}\right)e^x + (x + 2\sqrt{x})e^x \quad (1)$$

$$H(u) = (u - \sqrt{u})(u + \sqrt{u})$$

$$H'(u) = \left(1 - \frac{1}{2\sqrt{u}}\right)(u + \sqrt{u})$$

$$+ (u - \sqrt{u})\left(1 + \frac{1}{2\sqrt{u}}\right)$$



حل حلو



$$H(u) = u^2 - u$$

$$H'(u) = 2u - 1$$

$$J(v) = (v^3 - 2v)(v^{-4} + v^{-2})$$

$$J'(v) = (3v^2 - 2)(v^{-4} + v^{-2}) + (v^3 - 2v)(-4v^{-5} - 2v^{-3})$$



Text book questions



Differentiate

$$y = \frac{e^x}{x^2}$$

$$y' = \frac{x^2 e^x (1) - e^x (2x)}{(x^2)^2}$$

$$g(x) = \frac{3x - 1}{2x + 1}$$

$$g'(x) = \frac{(2x+1) 3 - (3x-1) 2}{(2x+1)^2}$$

$$y = \frac{e^x}{1+x}$$

$$y' = \frac{(1+x) e^x - e^x (1)}{(1+x)^2}$$

$$y = \frac{t^3 + 3t}{t^2 - 4t + 3}$$

$$y' = \frac{(t^2 - 4t + 3) (3t^2 + 3) - (t^3 + 3t) (2t - 4)}{(t^2 - 4t + 3)^2}$$



Text book questions



Differentiate

$$y = \frac{\sqrt{x}}{2+x}$$

$$y' = \frac{(2+x) \frac{1}{2\sqrt{x}} - \sqrt{x}}{(2+x)^2} \quad (1)$$

$$f(t) = \frac{\sqrt[3]{t}}{t-3}$$

$$= \frac{t^{\frac{1}{3}}}{t-3}$$

$$f'(t) = \frac{(t-3) \frac{1}{3} t^{-\frac{2}{3}} - t^{\frac{1}{3}}}{(t-3)^2} \quad (1)$$

$$y = \frac{A}{B + Ce^x}, \text{ where } A, B \text{ and } C \text{ are constants}$$

$$y' = \frac{(B+Ce^x) 0 - A(0+Ce^x)}{(B+Ce^x)^2} = \frac{-Ace^x}{(B+Ce^x)^2}$$

$$f(y) = \left(\frac{1}{y^2} - \frac{3}{y^4} \right) (y + 5y^3) = (y^{-2} - 3y^{-4})(y + 5y^3)$$

$$f'(y) = (-2y^{-3} + 12y^{-5})(y + 5y^3) + (y^{-2} - 3y^{-4})(1 + 15y^2)$$



Text book questions



Differentiate

$$y = \frac{v^3 - 2v\sqrt{v}}{v}$$

$$y = \frac{v^3 - 2v^{3/2}}{v}$$

$$y' = \frac{v(3v^2 - 2 \cdot \frac{3}{2}v^{1/2}) - (v^3 - 2v^{3/2})(1)}{v^2}$$



Tricky questions !!



$$f(x) = \frac{x^2 e^x}{x^2 + e^x}$$

$$\begin{aligned} f'(x) &= \frac{(x^2 + e^x)(x^2 e^x)' - x^2 e^x (x^2 + e^x)'}{(x^2 + e^x)^2} \\ &= \frac{(x^2 + e^x)(2xe^x + x^2 e^x) - x^2 e^x (2x + e^x)}{(x^2 + e^x)^2} \end{aligned}$$

$$V(t) = \frac{4+t}{te^t}$$

$$\begin{aligned} V'(t) &= \frac{te^t(4+t)' - (4+t)(te^t)'}{(te^t)^2} \\ &= \frac{te^t(1) - (4+t)(te^t + te^t)}{(te^t)^2} \end{aligned}$$

Let $f(x) = xe^x$

Find $f'(0)$

استفاد بشت + بشت استفاد

$$f'(x) = (1) e^x + x e^x (1)$$

$$\text{at } x=0 \rightarrow f'(0) = e^0 + 0 = 1$$

Let $f(x) = x^3 e^x$

Find $f'(2)$

$$f'(x) = 3x^2 e^x + x^3 e^x (1)$$

$$\begin{aligned} \text{at } x=2 \rightarrow f'(2) &= 3(4)e^2 + 8e^2 \\ &= 12e^2 + 8e^2 = 20e^2 \end{aligned}$$

Let $f(x) = \frac{x^2 - 1}{e^x}$

Find $f'(1)$

Mar 2018

$$f'(x) = \frac{e^x 2x - (x^2 - 1) e^x (1)}{(e^x)^2}$$

$$f'(1) = \frac{e^1 2(1) - 0}{(e^1)^2} = \frac{2e}{e^2} = \frac{2}{e}$$



Text book question



Find an equations of the tangent line and normal line to the given curve at the specified point.

$$y = 2xe^x, (0, 0)$$

$$y = 2x e^x$$

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$$y' = 2(1) e^x + 2x e^x$$

at (0,0)

$$y' = 2e^0 + 0 = 2(1) = 2$$

∴ The equation of tangent line :

$$y - y_0 = m(x - x_0)$$

$$y - 0 = 2(x - 0)$$

∴ The equation of the normal Line :

$$y - y_0 = m(x - x_0)$$

$$y - 0 = -\frac{1}{2}(x - 0)$$

Find an equation of the tangent line to the graph of the function

$$f(x) = \frac{e^x}{x+1} + x + 3 \quad \text{at } x = 0$$

Oct 2018

$$\begin{aligned} \text{at } x=0 &\longrightarrow f(0) = \frac{e^0}{0+1} + 0 + 3 \\ &= \frac{1}{1} + 3 = 4 \end{aligned}$$

$$\implies \therefore P(0, 4)$$

$$f'(x) = \frac{(x+1)e^x - e^x(1)}{(x+1)^2} + 1$$

$$\text{at } P(0, 4)$$

$$\begin{aligned} f'(0) &= \frac{(1)e^0 - e^0}{(1)^2} + 1 \\ &= \frac{1-1}{1} + 1 = 0 + 1 = 1 \end{aligned}$$

$\therefore$ The equation of the tangent line :

$$y - y_0 = m(x - x_0)$$

$$y - 4 = (1)(x - 0)$$



Text book question



Find equations of the tangent line and normal line to the given curve at the specified point.

$$y = \frac{2x}{x^2 + 1}, \quad (1, 1)$$

$$y' = \frac{(x^2 + 1) \cdot 2 - 2x \cdot (2x)}{(x^2 + 1)^2}$$

at $(1, 1)$

$$y' = \frac{(2) \cdot 2 - 2(2)}{(2)^2}$$

$$= \frac{4 - 4}{4} = \frac{0}{4} = 0$$

∴ The equation of the tangent line :

$$y - y_0 = m(x - x_0)$$

$$y - 1 = 0$$

∴ The equation of the normal line :

$$x = 1$$

Find the points on the graph

Apr 2017

$$y = \frac{x}{x^2 + 1}$$

Where the tangent line is horizontal.

$$y' = \frac{(x^2+1)(1) - x(2x)}{(x^2+1)^2}$$

$$= \frac{x^2+1 - 2x^2}{(x^2+1)^2}$$

$$= \frac{1 - x^2}{(x^2+1)^2}$$

∴ The tangent line is horizontal

$$∴ y' = 0$$

$$\begin{aligned} \rightarrow 1 - x^2 &= 0 \\ x^2 &= 1 \end{aligned} \quad \begin{aligned} &\rightarrow \boxed{x = -1} \\ &\rightarrow \boxed{x = 1} \end{aligned}$$

$$\text{at } x = -1 \rightarrow y = \frac{-1}{1+1} = -\frac{1}{2} \rightarrow P(-1, -\frac{1}{2})$$

$$\text{at } x = 1 \rightarrow y = \frac{1}{1+1} = \frac{1}{2} \rightarrow P(1, \frac{1}{2})$$

Find the point(s) on f where the tangent line is vertical

$$f(x) = \frac{x^{\frac{2}{3}}}{x-1}$$

$$f'(x) = \frac{(x-1)^{\frac{2}{3}} x^{-\frac{1}{3}} - x^{\frac{2}{3}} (1)}{(x-1)^2} \quad (1)$$

$$= \frac{\frac{2(x-1)}{3x^{\frac{1}{3}}} - x^{\frac{2}{3}}}{(x-1)^2}$$

$$= \frac{\frac{2(x-1)}{3x^{\frac{1}{3}}} - \frac{3x^{\frac{1}{3}} x^{\frac{2}{3}}}{3x^{\frac{1}{3}}}}{(x-1)^2} = \frac{\frac{2(x-1) - 3x}{3x^{\frac{1}{3}}}}{(x-1)^2}$$

$$= \frac{2x - 2 - 3x}{3x^{\frac{1}{3}} (x-1)^2} = \frac{-2 - x}{3x^{\frac{1}{3}} (x-1)^2}$$

$\Rightarrow \therefore$ Vertical tangent line $\rightarrow \therefore 3x^{\frac{1}{3}} (x-1)^2 = 0$

$$3x^{\frac{1}{3}} = 0$$

$$x = 0$$

$$(x-1)^2 = 0$$

$$x = 1$$

at $x=0$

1) $f(x)$ is continuous on $\mathbb{R} \setminus \{1\}$
 $\therefore f$ is continuous at $x=0$

$$2) \lim_{x \rightarrow 0} |f'(x)| = \infty$$

$$\text{at } x=0 \rightarrow y = \frac{0}{0-1} = 0$$

$\therefore$ at $P(0,0)$ there is V.T.L

at $x=1$

1) $f(x)$ is continuous on $\mathbb{R} \setminus \{1\}$
 $\therefore f$ is not continuous at $x=1$

$\therefore$ at $x=1$

No vertical tangent line.



الممنوعات



1) ممنوع أس سالب

2) ممنوع مقام غير موحد

Find the points on the tangent line of the curve $y = \frac{x-1}{x+1}$ that are parallel to the line $x - 2y = 2$

$$y = \frac{x-1}{x+1}$$

$$y'_1 = \frac{(x+1)(1) - (x-1)(1)}{(x+1)^2}$$

$$= \frac{x+1 - x+1}{(x+1)^2}$$

$$= \frac{2}{(x+1)^2}$$

$$x - 2y = 2$$

$$x - 2 = 2y$$

$$\div 2$$

$$y = \frac{1}{2}x - 1$$

$$y'_2 = \frac{1}{2}$$

$\therefore$ the tangent line of y is parallel to line

$$\therefore y'_1 = y'_2$$

$$\frac{2}{(x+1)^2} = \frac{1}{2}$$

$$4 = (x+1)^2$$

$$4 = x^2 + 2x + 1$$

$$x^2 + 2x - 3 = 0$$

$$(x-1)(x+3) = 0$$

$$x-1 = 0$$

$$x=1$$

$$x+3 = 0$$

$$x=-3$$

$$\rightarrow \text{at } x=1$$

$$y = \frac{1-1}{1+1} = \frac{0}{2} = 0$$

$$\Rightarrow (1, 0)$$

$$\text{at } x = -3$$

$$y = \frac{-3-1}{-3+1} = \frac{-4}{-2} = 2$$

$$\Rightarrow (-3, 2)$$



Text book questions



Find $f'(x)$ and $f''(x)$

$$f(x) = (x^3 + 1)e^x$$

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$$f'(x) = 3x^2 \cdot e^x + (x^3 + 1)e^x \quad (1)$$

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اشتقا بشتا + بشتا اشتقا

$$f''(x) = 6x e^x + 3x^2 e^x + 3x^2 e^x + (x^3 + 1)e^x$$

Find $f'(x)$ and $f''(x)$

$$f(x) = \sqrt{x}e^x$$

$$f'(x) = \frac{1}{2\sqrt{x}} e^x + \sqrt{x} e^x$$

$$= \frac{e^x}{2\sqrt{x}} + \sqrt{x} e^x$$

$$f''(x) = \frac{2\sqrt{x} e^x - e^x \cdot 2 \cdot \frac{1}{2\sqrt{x}}}{(2\sqrt{x})^2} + \frac{1}{2\sqrt{x}} e^x + \sqrt{x} e^x$$



Text book question



If $f(2) = 10$ and $f'(x) = x^2 f(x)$ for all x .

Find $f''(2)$

$$\text{at } x=2 \longrightarrow f'(2) = (2)^2 f(2) \\ = 4(10) = 40$$

$$\therefore f'(2) = 40$$

$$f'(x) = x^2 f(x)$$

$$f''(x) = 2x f(x) + x^2 f'(x)$$

at $x=2$

$$f''(2) = 2(2) f(2) + (2)^2 f'(2) \\ = 4(10) + 4(40)$$

$$= 40 + 160$$

$$= 200$$



Text book questions



If $g(x) = xf(x)$ where $f(3) = 4$ and $f'(3) = -2$.

Find an equation of the tangent line to the graph of g at the point where $x = 3$

at $x = 3$

$$g(3) = 3f(3) = 3(4) = 12$$

$$\Rightarrow \therefore P(3, 12)$$

$$g(x) = x \cdot f(x)$$

$$g'(x) = (1) f(x) + x f'(x)$$

at $x = 3$

$$\begin{aligned} g'(3) &= f(3) + 3f'(3) \\ &= 4 + 3(-2) = 4 - 6 = -2 \end{aligned}$$

$\therefore$ The equation of the tangent line :

$$y - y_0 = m(x - x_0)$$

$$y - 12 = -2(x - 3)$$



Tricky question !!



Find the 30th derivative of

$$f(x) = e^{3x}$$

$$f(x) = e^{3x}$$

$$f'(x) = 3e^{3x}$$

$$f''(x) = 3(3)e^{3x} = 3^2 e^{3x}$$

$$f'''(x) = 3^2 (3)e^{3x} = 3^3 e^{3x}$$

$$f^{(4)}(x) = 3^3 (3)e^{3x} = 3^4 e^{3x}$$

$$\Rightarrow \therefore f^{(n)}(x) = 3^n e^{3x}$$

$$\therefore f^{(30)}(x) = 3^{30} e^{3x}$$

Find the 30th derivative of

$$f(x) = xe^x$$

past exam

$$f(x) = xe^x$$

$$f'(x) = (1)e^x + xe^x = e^x(1+x)$$

$$f''(x) = e^x(1+x) + e^x(1) = e^x(1+x+1) = e^x(2+x)$$

$$f'''(x) = e^x(2+x) + e^x(1) = e^x(2+x+1) = e^x(3+x)$$

$$f^{(4)}(x) = e^x(3+x) + e^x(1) = e^x(3+x+1) = e^x(4+x)$$

$$\Rightarrow \therefore f^{(n)}(x) = e^x(n+x)$$

$$\therefore f^{(30)}(x) = e^x(30+x)$$